

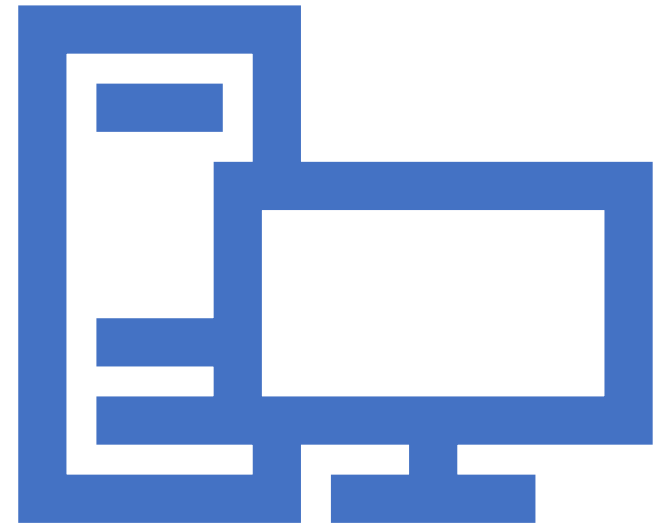
Introduction to Embedding

Natural Language Processing

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Input to ML models:

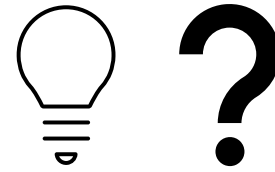
- Most machine learning algorithms can only take low-dimensional numerical data as inputs.



One-Hot-Encoding

- Corpus:
 - He likes apple.
 - She likes oranges.
- Vocabulary:
 - {He, likes, apple, she, oranges}
- One-hot vectors:

He-	[1,0,0,0,0]
likes-	[0,1,0,0,0]
apple-	[0,0,1,0,0]
she-	[0,0,0,1,0]
oranges-	[0,0,0,0,1]



What if the size of vocabulary is 300,000?

Does these vectors represent semantic similarity between apple and oranges?

- Input dimensions:

- W1 dimensions:

- W2 dimensions:

He- [1,0,0,0,0]

likes- [0,1,0,0,0]

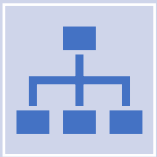
apple- [0,0,1,0,0]

she- [0,0,0,1,0]

oranges- [0,0,0,0,1]

- Input dimensions: #Training samples X #feature
- W1 dimensions: #features X Hidden size
- W2 dimensions: Hidden size X output size
- Note: For the sake of understanding and assignment we can generate vector for each sentence by adding the one-hot encode vectors of all of its words. In case of repetition, count it once.
- https://colab.research.google.com/drive/1CONh3rp2KYWexKimYbJLcU2Y_bSCS_4s?usp=sharing

Embeddings



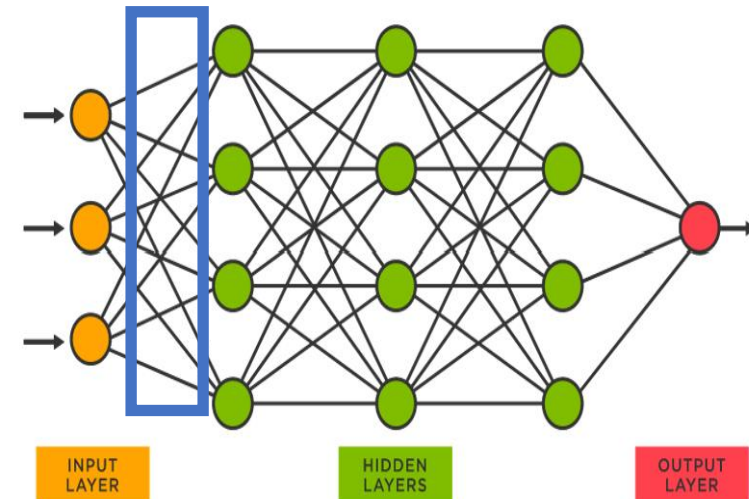
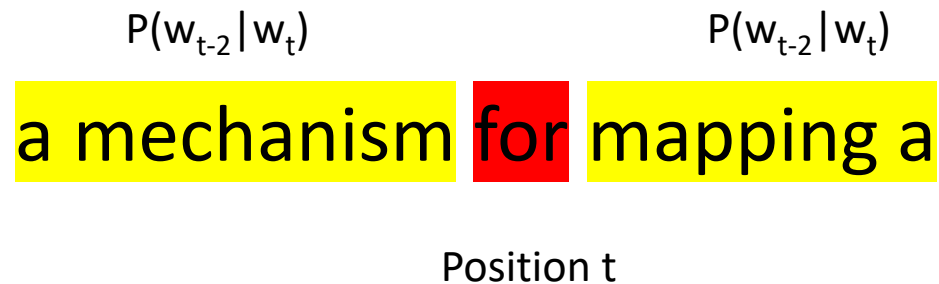
Embeddings are a mechanism for mapping a high dimensional space to a low-dimensional one while only retaining the most effective structural representations.



Embeddings are dense numerical representations of real-world objects and relationships, expressed as a vector.

Neural Network to Learn Embeddings

- Uses context to capture the semantic similarity.
 - E.g., maximize the probability of any context word given the current center word or vice versa.



Types of Embeddings



Word
embeddings.



Sentence
embeddings.



Document
embeddings.



Thank You!